

NONCOMPACTNESS OF THE FAMILY SO(3)-MONOPOLE SPACES

Introduction

We use the notation of [FS, §4], which is that of Feehan and Leness [FL2b]. Let $X = X(\Pi_M)$ be the closed manifold of [FS, §3] for the composites $\Pi_M = B \Pi_0 u^M (\phi_* B)^{p'}$ of [FS, Cor. 2.3], $u = (\phi_* B)^3 H B$, with the caps $C_\pm$ blown up once, exceptional classes $e_\pm$ and $\lambda_\pm = \langle \Lambda_0, e_\pm \rangle$ odd and large. It contains $n = 4M + O(1)$ copies of W_B , with three-spheres J_i and spheres S_i of square -4 , $[S_i] = F_{l,i} - F_{r,i}$, and $m = M + O(1)$ copies of W_H . The J_i cut X into *pieces*: negative pieces $X_{-2}(K) \cup_\phi (-X_2(K))$, positive pieces $P \cong \mathbb{CP}^2 \# 5\overline{\mathbb{CP}}^2$, and two end pieces containing the caps. The metrics g_t^R , $t \in \bar{Q} = [-1, 1]^n$, stretch the round J_i as $t_i \rightarrow -1$ and $\partial \nu S_i \cong L(4, 1)$, of positive scalar curvature, as $t_i \rightarrow +1$; the necks $[-R, R] \times Y_{\pm 2}$ outside the *blocks*, the copies of $W_B \cup W_H \cup W_B$, have a fixed length R . On a block the metric depends on (t_j, t_{j+1}) jointly, through the toric family of [MS, §6] (§5), not on each parameter separately as in [FS, §3]; the composition law [FS, Thm. 3.1] must be proved for this family.

Let t_e , $e \in \{0, 1\}^n$, have $w_2(t_e) \equiv w_e = w_0 + \sum_i e_i \text{PD}[S_i]$, $\Lambda_e = c_1(t_e)$, $\langle \Lambda_0, S_i \rangle = 2$ and $p_1(t_e) = -4\kappa$, so that $d_a = 8\kappa - 3(1 + b^+)$, $\Theta = \frac{1}{4}(\Lambda^2 - \sigma)$ and $n_a = \Theta - \kappa$ do not depend on e . Represent $z = \mu(S_1) \cdots \mu(S_n)z'$, $\deg z = d_a + n$, by $\mathcal{V}(z)$, the intersection of the jumping-line divisors V_{S_i} with representatives of the cap classes z' , and μ_c by $n_a - 1$ sections of $\mathbb{L}_t$ [FL2b, (3.10), (3.12)] that sample the spinor on every piece, the μ_c -sections. Then $\mathcal{Z}_e = \mathcal{V}(z) \cap \mathcal{W}^{n_a-1} \cap \mathcal{M}_{t_e}^{*,0}(Q)/S^1$ [FL2b, (3.62)] has dimension $(d_a + 2n_a + n - 1) - (d_a + n) - 2(n_a - 1) = 1$, and [FS, Thm. 4.1] gives $2^{n_a-1}\Omega = 0$, $\Omega = \sum_e \epsilon(e) \#(\bar{\mathcal{V}}(z) \cap \bar{M}_\kappa^{w_e}(Q))$, once the closure of $\mathcal{Z}_e$ over $\bar{Q}$ has no ends but those at the link $\{\|\Phi\|_{L^2}^2 = \varepsilon_0\}$ of the instanton stratum and ends over the faces $\{t_i = +1\}$ that cancel in $\sum_e \epsilon(e)$. Sections 1–6 list the strata of this closure at a fixed $R \geq \max(R_0(M), T_0(M))$, $R_0(M)$ from the composition law and $T_0(M)$ from §4; §7 revises (H5).

Over a face on which k three-spheres and l lens spaces are stretched, an ideal limit [FL1, Def. 4.19, Thm. 4.20] has a main component on each of the pieces closed up by balls, $\hat{\Gamma}_0 \supset C_-, \dots, \hat{\Gamma}_k \supset C_+$, with points of concentration, trajectories on the stretched necks, and connections on the caps $\hat{\nu}_i$ of the lens spaces. By the proof of [FL2a, Prop. 3.1], which is local, a main component lies in $\mathcal{M}^{*,0}$ (irreducible, $\Phi \neq 0$), or is *anti-self-dual* (irreducible, $\Phi \equiv 0$), *Seiberg–Witten* (a point of $\iota(M_\mathfrak{s}) \times \text{Sym}^\ell$ of its piece) or an *abelian instanton* (reducible, $\Phi \equiv 0$). At a reducible, $E = L_1 \oplus L_2$, $v = c_1(L_1) - c_1(L_2) \equiv w$, $c_1(\mathfrak{s}) = \Lambda + v$ and $\ell = \kappa + \frac{1}{4}v^2$ [FL2b, Lemma 3.32, (3.64)]. A component on $\hat{\Gamma}_j$ that retains p_j parameters and carries the constraints V_{S_i} of s_j spheres and cap classes of degree z_j has cut-down anti-self-dual and monopole indices $q_j = d_a(\hat{\Gamma}_j) + p_j - 2s_j - z_j$ and $i_j = q_j + 2n_a(\hat{\Gamma}_j)$, lens ends being filled by the reducibles of §3 and the energy of trajectories and points of concentration counted in its κ . A *chain* is a maximal set of consecutive reducible components, and $\delta(\Gamma) = \sum_{j \in \Gamma} (i_j + 4)$. Claims are labelled VERIFIED (proved here, or checked against the source or by an exact computation), PLAUSIBLE or SPECULATIVE.

1. Interior parameters

Proposition 1.1. *With the choices of (H5a) made before R , every solution has $|\Phi|^2 \leq K_3$ and $|F_A^{0,+}| \leq C|\Phi|^2 + |P_+|$, P_+ the perturbation of the curvature equation, with K_3, C independent of $t \in \bar{Q}$ and R . If $\zeta = f^2$, $0 \leq f \leq 1$, has compact support, then $\frac{1}{2} \int \zeta^2 |\nabla_A \Phi|^2 + \frac{1}{16} \int \zeta^2 |\Phi|^4 \leq 2 \int \zeta^2 |P_D|^2 + 4 \int \zeta^2 (h^2 + |P_+|^2) + 36 \int |d\zeta|^4 \zeta^{-2}$, with P_D the perturbation of the Dirac equation and h the negative part of the least eigenvalue of $\frac{1}{4}\text{Scal} + \frac{1}{2}\rho(F_{A\Lambda}^+)$; and $h = 0$ on the round and lens necks, $\int h^2 = o(\epsilon^{-1})$ on the tubes of §5.*

Sketch. [FL1, Lemma 4.4, Rem. 4.6], Floer’s holonomy perturbation entering through its sup norm; then pair $D_A^* D_A \Phi = D_A^* P_D$ with $\zeta^2 \Phi$, and use [FL1, Lemma 4.1], $\langle (\Phi \Phi^*)_{00} \Phi, \Phi \rangle = \frac{1}{4}(s^4 + t^4) + \frac{5}{2}s^2 t^2 \geq \frac{1}{4}|\Phi|^4$ (s, t the singular values of Φ) and Young’s inequality. $\square$

Proposition 1.2. *Assume (H5). Over $\text{int } Q$ the closure of $\mathcal{Z}_e$ consists of $\mathcal{Z}_e$ and the finitely many points of $\mathcal{V}(z) \cap M_\kappa^{w_e}(Q)$, near each of which $\mathcal{Z}_e$ has 2^{n_a-1} ends, counted with the sign of the instanton for $o(\Omega, w_e)$ and a sign fixed by conventions. Precisely: (a) strata at level $\ell \geq 1$ with $\Phi \neq 0$ have expected dimension at most $1 - 2\ell$; (b) those at level $\ell \geq 1$ with $\Phi \equiv 0$, at most -4ℓ ; (c) no Seiberg–Witten stratum occurs (§5); (d) an abelian instanton would contain both caps, which (H5c) excludes.*

Sketch. A limit is an ideal monopole in $\mathcal{M}_{t(\ell)}(g_t^R) \times \text{Sym}^\ell(X)$ [FL1, Thm. 4.20], [FL2a, Thm. 2.12]. (a) At level ℓ , $\dim \mathcal{M}_{t(\ell)}^{*,0}(Q)/S^1 = d_a + 2n_a + n - 1 - 6\ell$ [FL2b, (3.20)]; constraints whose support meets no point of concentration persist, and a point releases constraints of degree at most 4 minus the dimension of its allowed positions [FL2b, (3.25)], e.g. V_{S_i} , of degree two, at a point moving on S_i . So the conditions left have degree at least $d_a + n + 2(n_a - 1) - 4\ell$. Without (H5d) a point anywhere in νS_i could release V_{S_i} , and level one would give $1 - 6 + 6 = 1$. (b) The background lies in $M_{\kappa-\ell}^{w_e}(Q)$, of dimension $d_a + n - 8\ell$, under conditions of degree at least $d_a + n - 4\ell$. At level 0 a limit with $\Phi \equiv 0$ lies in $\mathcal{V}(z) \cap \iota(M_\kappa^{w_e}(Q))$ [FL2b, Lemma 3.15], and on the link, as $|\Phi|$ is bounded and does not concentrate. The link is [FL2b, Lemmas 3.22–3.28, Prop. 3.29] over Q , the terms of the obstruction map involving t being cubic on

$\mathcal{V}(z): \langle (2h)^{n_a-1} h^c, [\mathbb{CP}^{n_a+c-1}] \rangle = 2^{n_a-1}$, with a sign for each e whose dependence on e is the comparison of §3. $\square$

Status. VERIFIED: the counts, and the failure of (a) without persistence. PLAUSIBLE: compactness and transversality in families (90%), (H5d) (85%), the link over Q (85%). Confidence for the interior: 80%.

2. Faces broken along the three-spheres J_i

Lemma 2.1. *On the round S^3 with flat background, $(\theta, 0)$ is the only solution of the three-dimensional monopole equations; it has stabilizer $SU(2)$, $H^1 = 0$ and invertible twisted Dirac operator. A finite-energy solution on $\mathbb{R} \times S^3$ has $\Phi \equiv 0$ and is anti-self-dual of charge $c \in \mathbb{Z}$, index $8c - 3$ and Dirac index $-c$, with zero kernel. For an ideal limit over a face with k stretched three-spheres,*

$$(2.1) \quad \sum_{j=0}^k (q_j + 4) = 4, \quad \sum_{j=0}^k i_j - 2(n_a - 1) = 2 - 4k.$$

Proof. Integrating [FL1, Lemma 4.1] gives $0 = \|\nabla \Psi\|^2 + \int \frac{1}{4} \text{Scal} |\Psi|^2 + \|(\Psi \Psi^*)_{00}\|^2$, so $\Psi = 0$, and $\pi_1(S^3) = 1$. For (2.1): $\sum_j d_a(\hat{\Gamma}_j) = d_a - 3k$, each stretched J_i contributing $h^0(\theta) = 3$; $\sum_j n_a(\hat{\Gamma}_j) = n_a$, the Dirac operator of S^3 being invertible; $\sum_j p_j = n - k$, $\sum_j s_j = n$ and $\sum_j z_j = \deg z' = d_a - n$. $\square$

So each stretched three-sphere lowers the dimension by four, three for the stabilizer of $(\theta, 0)$ and one for t_i . A gauge-invariant f with $f(\zeta x) = \zeta^2 f(x)$ vanishes at anti-self-dual, Seiberg–Witten and abelian components, as every scalar fixes $(A, 0)$ and $\text{diag}(\zeta, \zeta^{-1})$ fixes $A_{L_1} \oplus A_{L_2}$ and multiplies Φ by ζ ; so at a broken limit the μ_c -sections restrict to the components in $\mathcal{M}^{*,0}$.

Proposition 2.2. *Let the perturbations near a limit act on one piece at a time, and let F be the set of its components in $\mathcal{M}^{*,0}$, $f = |F| \geq 1$. The limit lies in $\mathcal{Z}_F \times \prod_{j \notin F} M_j$, where $\mathcal{Z}_F$ is the space of the components in F , cut down by their constraints and all μ_c -sections, modulo the diagonal circle; if $\mathcal{Z}_F$ is regular,*

$$(2.2) \quad \dim \mathcal{Z}_F = D_F = 5 - 4f - \sum_{j \notin F} (i_j + 4) - \lambda_F,$$

where $\lambda_F \geq 0$ is the codimension of the lens ends (at least 1 each), trajectories (at least 4 each) and points of concentration (at least $2a$ for charge a) on the components in F . If $D_F < 0$ the limit does not exist.

Proof. $D_F = \sum_{j \in F} i_j - \lambda_F - 1 - 2(n_a - 1)$; apply (2.1) and $k + 1 = f + \#\{j \notin F\}$. $\square$

The components outside F enter only through the topological numbers i_j , and nothing analytic is asked of them. Regularity of the whole broken configuration cannot be had: a trajectory of charge c has a Dirac cokernel of dimension c on a neck where every perturbation vanishes. An anti-self-dual component is of low index if $i \leq -3$, or if it contains a cap, $k \geq 2$ and $i \leq -1$; a Seiberg–Witten component containing a cap is of low index if its chain has $\delta \leq 1$.

Theorem 2.3. *Assume (H5), and let ζ be a limit of points of $\mathcal{Z}_e$ over a face with $k \geq 1$ stretched three-spheres. (a) If ζ has no component in $\mathcal{M}^{*,0}$, it does not exist. (b) If ζ has a component in $\mathcal{M}^{*,0}$ and none of low index, then $D_F \leq -1$, and ζ does not exist; the thresholds are sharp. (c) Components of low index lie in two unions of pieces fixed before M , the end regions: the pieces within about $8\lambda_{\pm}^2 + 5p'$ copies of W_B of $C_{\pm}$ for anti-self-dual components, within about $\frac{1}{2}\lambda_{\pm}^2$ for Seiberg–Witten ones, together with $B\Pi_0$ and $(\phi_* B)^{p'}$.*

Sketch. (a) In (2.1) anti-self-dual components have $q + 4 \geq 4$ by (H5e), Seiberg–Witten components containing a cap have $q + 4 \geq 8$ (Proposition 5.4), and abelian instantons contain no cap. So the $N_0 \geq 2$ components of the first two kinds are separated by at most $N_0 - 1$ chains without a cap, each with $\sum (q_j + 4) \geq -2$ (§5), and $\sum_j (q_j + 4) \geq 2N_0 + 2 \geq 6$. (b) The components outside F form inner intervals, between two members of F , at most $f - 1$ of them, and outer intervals, between a cap and the nearest member of F . In an inner interval, chains (with $\delta \geq -2$, §§5–6) alternate with anti-self-dual components ($i + 4 \geq 2$), so it contributes at least -2 ; an outer interval begins with an anti-self-dual cap component ($i + 4 \geq 4$) or a Seiberg–Witten one with $\delta \geq 2$, and contributes at least 2. So $D_F \leq 5 - 4f + 2(f - 1) - 2\#\{\text{outer}\} \leq -1$; for $k = 1$, $D_F = -3 - i_c$. A cap anti-self-dual component of index -1 next to a chain with $\delta = -2$, or at $k = 1$ one of index -3 , gives $D_F = 0$. (c) For an anti-self-dual component, $q \geq 0$ and $n_a \in \mathbb{Z}$ give $i \geq Q - 6\lfloor Q/8 \rfloor$, $Q = 8\Theta(\hat{\Gamma}) - 3(1 + b^+) + p - 2s - z$, where $\Theta = 0$ on a copy of W_B , $\Theta(W_H) = 1$, and Θ has a term $\frac{1}{4}(1 - \lambda_{\pm}^2)$ at a cap; so $i \geq -2$ inside u^M , and $i \leq -3$ needs five consecutive negative pieces or a cap. For Seiberg–Witten components see Proposition 5.4. $\square$

So at $k = 1$, where both pieces contain a cap, a Seiberg–Witten component next to one in $\mathcal{M}^{*,0}$ is excluded only if it is short. At the node, [FS, Lemma 3.2] holds verbatim for monopoles, as V_{S_i} depends only on $A|_{S_i}$: a limit satisfies the constraint of a capped half of S_i , or has a point of concentration on S_i or a trajectory with a jumping line on $\mathbb{R} \times K$.

Proposition 2.4. *At a limit with an anti-self-dual component A_r on $\hat{\Gamma}_r$ and components in $\mathcal{M}^{*,0}$ elsewhere, a single-valued $\mathcal{G}_r$ -equivariant term in the Dirac equation on $\hat{\Gamma}_r$ depending on A_r and on other pieces vanishes, as $-1 \in \mathcal{G}_r$ fixes its arguments and negates its value; terms linear in Φ_r are compact; and terms comparing frames across the neck*

bring in the gluing parameter, with ends over the face that nothing controls. With the terms $\pm\beta(|s_W|)s_W^{1/2}g_{W,\Gamma}(A|_\Gamma)$ of (H5f), s_W of weight two in the spinor on a piece $W \not\subset \Gamma$, the Dirac equation at a stratum M_Γ of low index is solvable only where a section of a bundle of rank $-2n_a(\hat{\Gamma}) > \dim M_\Gamma$ vanishes, a set of dimension $i < 0$; and if $K \supset F$ adds the components of low index, $5 - 4|K| - \sum_{j \notin K}(i_j + 4) \leq -1$.

Status. VERIFIED: Lemma 2.1; the vanishing of the μ_c -sections; (2.2); Theorem 2.3, exhaustively over arrangements of up to ten components, and with (a) the exclusion part of the finiteness hypothesis of [FS, Thm. 4.1]; Proposition 2.4; the algebra at the node. PLAUSIBLE: additivity across S^3 ; (H5e) (85%); the linear gluing at the node (85%); (H5f), new (55–60%). Confidence: 45%.

3. Faces broken along the lens spaces $\partial\nu S_i$

Lemma 3.1. *On $L(4, 1)$ the flat $SU(2)$ connections are $\pm\theta$, with stabilizer $SU(2)$, and $\gamma = \text{diag}(i, -i)$, with stabilizer $U(1)$; all have $H^1 = 0$ and invertible Dirac operators. The metric $dr^2 + A^2(\sigma_1^2 + \sigma_2^2) + B^2 \tanh^2(4r/B)\sigma_3^2$, $B < 2A$, on the cap $\hat{\nu}_i$ has scalar curvature $64B^{-2}\text{sech}^2(4r/B) + 8A^{-2} - 2B^2 \tanh^2(4r/B)A^{-4} > 0$, so with an anti-self-dual background no finite-energy spinor survives on $\hat{\nu}_i$. The reducibles on $\hat{\nu}_i$ have $v = 2j\xi$, $\xi(S_i) = 1$, energy $E = j^2/4$ and restriction $\mathcal{O}(j) \oplus \mathcal{O}(-j)$ to S_i , so lie in V_{S_i} iff $j \neq 0$, and limit γ for j odd, $\pm\theta$ for j even (θ and $-\theta$ exchanged between $e_i = 0, 1$). Their framed indices are $8E$, and $6E$ at $\pm\theta$, $2 + 6(E - \frac{1}{4})$ at γ for monopoles, the complex Dirac index being $-E$, resp. $\frac{1}{4} - E$. For $|j| = 1$ the Dirac operators of the two lifts, of evaluations $(4, 0)$ and $(0, -4)$, are invertible.*

Proposition 3.2. *Assume (H5). Over an open face $\{t_i = +1\}$ the closure of $\mathcal{Z}_e$ consists of finitely many ends, each converging to a rigid solution in $\mathcal{M}^{*,0}$ on $X \setminus \nu S_i$ with limit γ , glued to the reducible of energy $\frac{1}{4}$ on $\hat{\nu}_i$, one end for each such exterior solution.*

Sketch. A cap of monopole framed index d_c carries V_{S_i} , of degree two, and the face removes t_i , so an exterior component in $\mathcal{M}^{*,0}$ has dimension $2 - d_c$: 0 at energy $\frac{1}{4}$, $-6h$ at energy $\frac{1}{4} + h$ at γ , at most -4 at $\pm\theta$, where the least energy in V_{S_i} is 1, and at most $-2a$ with points of concentration of charge a . An irreducible cap of energy $\frac{1}{4}$ in V_{S_i} has framed index 2 and a free action of $\text{Stab}(\gamma)/\{\pm 1\}$, so is absent for generic data. An anti-self-dual exterior gives $1 - 8E \leq -1$, a Seiberg–Witten one is excluded by Theorem 5.3, and an abelian one contains both caps. The gluing parameter lies in $\text{Stab}(\gamma)/(\text{Stab } x \cdot \text{Stab } c)$, a point, and V_{S_i} cuts the normal line of the cap transversely, the non-split extension of $\mathcal{O}(1)$ by $\mathcal{O}(-1)$ being $\mathcal{O} \oplus \mathcal{O}$. $\square$

No coupled perturbation is needed here, no limit of the cut-down instanton spaces lies here, and at a corner two lens ends give $1 - 2 = -1$; so lens ends lie over open faces only, and cancel face by face:

Theorem 3.3. *Let $e_i = 0$ and $e' = e + e_i$. The exterior problems for e and e' on $X \setminus \nu S_i$ coincide, and for each exterior solution x the ends of $\mathcal{Z}_e$ and $\mathcal{Z}_{e'}$ at x have signs with $\sigma_{e'}(x) = \epsilon \sigma_e(x)$, ϵ the sign of [FS, Thm. 1.1]; so they cancel in $\sum_e \epsilon(e)$.*

Sketch. The orientation is $o_e \otimes (\text{the complex orientation of } \det D_{A,\theta})$ [FL2b, Def. 2.3, (2.5)], o_e glued along the necks along $Y_{\pm 2}$ from the caps and the copies of W_H and W_B . As $\text{PD}[S_i]$ vanishes on $X^\circ = X \setminus \nu S_i$ and $H^3(X^\circ) = 0$, $E_e|_{X^\circ} \cong E_{e'}|_{X^\circ}$ canonically up to homotopy, and excision splits the determinant lines into a common exterior factor and cap factors. Tensoring by $L_{-2\xi}$ on $\hat{\nu}_i$, followed by a Weyl conjugation at γ , preserves $\text{ad } E$ and V_{S_i} with its complex coorientation; it does not act on the spinors, but both cap Dirac operators are invertible and the exterior one is common, so the spinor contributes $+1$. What remains is the comparison of o_c and $o_{c+\text{PD}(S)}$ at the lens face of W_B , which is ϵ by (H4). No line bundle on X relates E_e and $E_{e'}$: their w_2 differ by $\text{PD}[S_i]$, odd on a class of an adjacent piece. $\square$

Status. VERIFIED: Lemma 3.1 (scalar curvature from the Christoffel symbols; Dirac indices by excision against $\mathbb{F}_4$ and $\mathbb{P}(1, 1, 4)$ and by η -invariants); the counts of Proposition 3.2; Theorem 3.3 given excision and (H4). PLAUSIBLE: gluing at the reducible of energy $\frac{1}{4}$ (80%); excision with stabilizer $U(1)$ (90%); (H4) (80%); the cap metrics (90%). Confidence: 75%.

4. The long necks along $Y_{\pm 2}$

At a fixed R no neck along $Y_{\pm 2}$ is stretched and no stratum is broken there; the necks matter through the constants.

Proposition 4.1. *Let $g(K) \geq 2$. (a) $Y_{\pm 2}$ is not an L -space. (b) For every metric and small perturbation with nondegenerate reducibles, the three-dimensional Seiberg–Witten equations on $Y_{\pm 2}$ have an irreducible solution. (c) $Y_{\pm 2}$ has no metric of nonnegative scalar curvature; so $\int_{[-R,R] \times Y_{\pm 2}} h^2 \geq cR$, and a solution near a critical point with nonzero spinor along an interval of length L has $\int |\Phi|^4 \geq cL$.*

Proof. (a) Otherwise K would be an L -space knot, and the integer surgery formula [OS] would give $\text{rank } \widehat{HF}(S_2^3(K)) = 2 + 2(2g - 3) > 2$ (the rational surgery bound gives only $g \leq 2$); and $S_{-2}^3(K) = -S_2^3(\bar{K})$. (b) Otherwise monopole

Floer homology, isomorphic to Heegaard Floer homology, would have rank $|H_1| = 2$. (c) Such a metric, if not flat, admits no irreducible solution, and no flat orientable three-manifold has $H_1 = \mathbb{Z}/2$. $\square$

So the bounds of Proposition 1.1 grow with R ; what stays bounded is the drop of the Chern–Simons–Dirac functional. In temporal gauge on a neck write the equations as $\dot{\Phi} + D_a \Phi = e$, $\dot{a} + *F_a + \sigma \nabla \mathcal{W}(a) = q(\Phi) + r$, $\langle q(\Phi), b \rangle = -c \langle \rho(b) \Phi, \Phi \rangle$, $c > 0$, with $\mathcal{W}$ the holonomy perturbation, $|\mathcal{W}| \leq M_{\mathcal{W}}$, $\sigma = \pm 1$, and r, e the other perturbations, of total squared L^2 norm ν .

Lemma 4.2. *With $H(t) = \langle D_{a(t)} \Phi(t), \Phi(t) \rangle_{L^2(Y)}$ and κ normalized by $\|F^0\|^2 = 2\|F^{0,+}\|^2 + 8\pi^2 \kappa$,*

$$4\pi^2 \kappa_{[u,v] \times Y} = \int_u^v (\|\dot{a}\|^2 + 2c\|\dot{\Phi}\|^2 - \langle r, \dot{a} \rangle - 2c\langle e, \dot{\Phi} \rangle) dt + c(H(v) - H(u)) + \sigma(\mathcal{W}(a(v)) - \mathcal{W}(a(u))).$$

Hence every neck middle has $\kappa \geq -C_Y(1 + \nu)$, and every component V of X minus the neck middles has $\kappa_V \leq \kappa + C(1 + M)$ and $\|F_A^0\|_{L^2(V)}^2 \leq C(1 + \kappa + M)$, with C_Y, C independent of R and t ; an Uhlenbeck limit along any sequence (t_ν, R_ν) has at most $C(1 + \kappa + M)$ points of concentration.

Proof. In temporal gauge $4\pi^2 \kappa_{[u,v]} = -\int (*F_a, \dot{a})$; substitute the curvature equation and $H' = \langle \rho(\dot{a}) \Phi, \Phi \rangle - 2\|\dot{\Phi}\|^2 + 2\langle e, \dot{\Phi} \rangle$. Choose u, v in unit strips at the ends of the neck where $|H| \leq B_Y(1 + \nu)^{3/4}$, as Proposition 1.1 allows; the rest follows from additivity of κ and Proposition 1.1 on the components. $\square$

By Lemma 4.2 the inputs of (H5) that must not depend on R (the cap constant C_W , the absence of abelian instantons containing a cap, and Theorem 5.2(f)) can be fixed before R , as the composition law requires: each is proved by contradiction along sequences with $R \rightarrow \infty$, on completed caps and toric pieces, where the critical points at the ends play no role. $R = \infty$ is not a face of the problem.

Status. VERIFIED: Lemma 4.2, also numerically on a finite-dimensional model (relative error $5 \cdot 10^{-10}$; for $c < 0$ the integrand has no sign), and its consequences as deductions; Proposition 4.1, given the surgery formula and the isomorphism of Floer theories. PLAUSIBLE: the inputs fixed before R (70%); compactness as $R \rightarrow \infty$ in their proofs (80%). Confidence: 70%.

5. Seiberg–Witten strata on X , on the positive pieces and on the caps

A reducible pair on a piece with $b_1 = 0$ and $w_2(\mathfrak{t}) \neq 0$ solves the Seiberg–Witten equations of $c_1(\mathfrak{s}) = \Lambda + v$ with $\eta = F_{A_\Lambda}^+ [\text{FL2a, Lemmas 3.12, 3.13}]$; its stratum $\iota(M_{\mathfrak{s}}) \times \text{Sym}^\ell$ has tangential index $d_s = \frac{1}{4}(c_1(\mathfrak{s})^2 - 2\chi - 3\sigma) = n_a + \ell + \frac{1}{2}\Lambda \cdot v - (1 + b^+)$ [FL2a, (2.63)], the normal part completing it to $d_a + 2n_a + 6\ell$. The μ_c -sections vanish there (§2), so these strata are excluded by level on X and by (2.2) at broken limits. There are no twisted reducibles, as $H_1 = 0$ on $X, X \setminus \nu S_i$ and the pieces.

Incidence. If points of $\mathcal{Z}_e$ converge, off the faces $\{t_i = -1\}$, to a Seiberg–Witten point of class v , each S_i with $\langle v, S_i \rangle = 0$ carries a point of concentration, by (H5d) and as the splitting type on S_i is $| \langle v, S_i \rangle | / 2$; so $\ell \geq T(v) = \#\{i : \langle v, S_i \rangle = 0\}$. On a positive piece P_j between J_j and J_{j+1} let $h_1 = \langle v, F_{l,j} \rangle$ and $h_2 = \langle v, F_{r,j+1} \rangle$ be the evaluations on the halves adjacent to P_j in the neighbouring pieces, and $Q_I = v_{P_j}^2 - \frac{1}{2} \sum_{i \in I} h_i^2$ for $I \subset \{1, 2\}$.

Proposition 5.1. *Suppose that on every positive piece $Q_I \leq -2$ for some I , or $Q_I \leq 2$ with one of $V_{S_j}, V_{S_{j+1}}$ not satisfied by v ; that $v_{\pm}^2 \leq C_{W_{\pm}}$ on the caps; and that positive pieces are at distance at least two. Then $-\frac{1}{4}v^2 + T(v) \geq \frac{1}{2}(n - m) - C$, $C = \frac{1}{4}(C_{W_-} + C_{W_+})$.*

Proof. $H^2(X; \mathbb{Z})$ splits orthogonally along the J_i . A negative piece contributes half the sum of the squares of its two even half evaluations to $-v^2$. Each positive piece, with the halves in I , contributes at least 2 to $-v^2 + 4T(v)$; each of the $n - 2m$ spheres not adjacent to a positive piece lies in V_{S_i} through a nonzero half, contributing at least 2 to $-v^2$, or adds 1 to $T(v)$; at distance two no half is used twice. $\square$

It is sharp on nine chains, and fails at distance one. It replaces [FS, Prop. 4.2], which is true but does not suffice: on P , $(\pi v)_P^2$ is unbounded on the classes allowed below (80, 360, 1520, ...).

Theorem 5.2. *Let the metric on the plumbing of S_j , the fibre class U of P_j and S_{j+1} , a chain of spheres of squares $-4, 0, -4$, be conformal to the toric Kähler metric of [MS, §6], with period $\omega = U + aX_1 + bX_2$, $X_1 = -S_j$, $X_2 = S_{j+1}$, $0 \leq a, b \leq \frac{1}{4}$; so $\omega = \sum_I \theta_I H_I$, $\theta_I \geq 0$, $H_I = U + \frac{1}{4} \sum_{i \in I} X_i$, and $\Lambda \cdot \omega < 0$. (a) On a complete piece with $\text{Scal} \geq 0$, positive somewhere, $b_{L^2}^+ = 1$, period ω , ends of positive scalar curvature and harmonic background, a Seiberg–Witten solution has $(v \cdot \omega)(c_1(\mathfrak{s}) \cdot \omega) < 0$, and an abelian instanton has $v \cdot \omega = 0$. (b) The toric metrics $\frac{1}{2}M_{ij}d\mu_i d\mu_j + 2(M^{-1})_{ij}d\vartheta_i d\vartheta_j$, $M = \text{diag}(c, 0) + \sum_\nu n_\nu n_\nu^t / l_\nu$, $c \geq 0$, have $\text{Scal} \geq 0$. (c) If $\omega \neq U$ and $v \cdot U \neq 0$, a reduced class satisfies the hypothesis of Proposition 5.1 for some $I \neq \emptyset$ with $\theta_I > 0$; if $v \cdot U = 0$, then $v_P^2 \leq -2$. (d) Near the corner $t_j = t_{j+1} = -1$, ω is forced to U ; there the block contains a tube $S^2 \times S^1 \times [0, L]$, $L \geq c/\epsilon$, on which $4\pi^2(v \cdot U)^2 L \leq 2\|D^+\|^2 + 16\pi^2 \kappa_V$, so $v \cdot U = 0$ for small ϵ . (e) On the isolated P , for any metric of nonnegative scalar curvature, positive somewhere, every Seiberg–Witten class with $d_s = \frac{1}{4}(c_1(\mathfrak{s})^2 - 4) \geq 0$ has $v_P^2 \leq -2$. (f) For every*

$t \in \bar{Q}$ and $R \geq T_0(M)$, every reducible component containing P_j of a limit of points of $\mathcal{Z}_e$ satisfies the conclusion of (c), or $v \cdot U = 0$.

Sketch. (a) Integrating the Weitzenböck formula gives $0 = \int (|\nabla \psi|^2 + \frac{1}{4} \text{Scal} |\psi|^2) + 8\pi^2 (v \cdot \omega)(c_1(\mathfrak{s}) \cdot \omega)/\omega^2 + 2\|D_\perp^+\|^2$, D the curvature of the reduction. (b) By Abreu's formula [Ab], $\text{Scal} = -2\partial_i \partial_j B_{ij}$, $B = M^{-1}$, and $-\partial_i \partial_j B_{ij} = q^t(I - P)q + z^t((I - P) \circ P \circ P)z$ with $P_{\nu\mu} = n_\nu^t B n_\mu / \sqrt{l_\nu l_\mu}$, $0 \leq P \leq I$, $z_\nu = l_\nu^{-1/2}$, $q_\nu = z_\nu P_{\nu\nu}$; apply Schur's product theorem. (c) By (a), $\Lambda \cdot \omega < c_1(\mathfrak{s}) \cdot \omega < 0$, and convexity gives $\theta_I > 0$ with $\text{sign}(u) v \cdot H_I < |\Lambda \cdot H_I|$, $u = v \cdot U$; identities such as $Q_{\{1\}} = 8(v \cdot H_{\{1\}})u - 4u^2 - \sum_b (v \cdot T_b - u)^2 - \frac{1}{2}(h_1 - 2u)^2$, T_b the exceptional classes, and parity give the bounds. (d) $\int_{S^2} D = 2\pi v \cdot U$ on fibres of area 2π . (e) If $v_P^2 \geq 0$, then v is causal and $c_1(\mathfrak{s})$ timelike, in opposite half-cones by (a), so $c_1(\mathfrak{s}) \cdot v \leq 0$ and $\Lambda_P^2 = c_1(\mathfrak{s})^2 - 2c_1(\mathfrak{s}) \cdot v + v^2 \geq 4$, whereas $\Lambda_P^2 \leq 2$ for every (e_j, e_{j+1}) . (f) By contradiction, letting the regularization tend to zero and the isolation lengths and R to infinity, with Lemma 4.2: a limit is an unperturbed reduction on the completed toric piece, contrary to (a)–(d). $\square$

Theorem 5.3. Assume (H5c). For all large M no Seiberg–Witten stratum of X , at any level, meets the closure of $\mathcal{Z}_e$ over $\text{int } Q$ or over the faces $\{t_i = +1\}$.

Proof. Incidence and Proposition 5.1 give $\kappa = \ell - \frac{1}{4}v^2 \geq \frac{1}{2}(n - m) - C$. As $8\kappa = n + 3m + c_\kappa$, $c_\kappa = \deg z' + 3 + 3b^+(C_- \sqcup C_+)$, this says $7m - 3n + c_\kappa + 8C \geq 0$, which is false for $n = 4M + O(1)$, $m = M + O(1)$ and M large. $\square$

Proposition 5.4. (a) No abelian instanton contains a cap $C = C_\pm$, and Seiberg–Witten components have $v_C^2, c_1(\mathfrak{s})_C^2 \leq C_W$, with C_W depending only on C . (b) A Seiberg–Witten component on an end piece made of fewer than L_* pieces, regular in its p -parameter family, has $d_s + p \geq 0$, hence $|c_1(\mathfrak{s}) \cdot e_\pm| \leq B$ independently of $\lambda_\pm$, $|v \cdot e_\pm| \geq \lambda_\pm - B$, and $q + 4 \geq 8$, $\delta \geq 2$ for $\lambda_\pm$ large. (c) On a longer end piece $q \geq 4$, but i can be as low as about $-\frac{1}{8}\lambda_\pm^2$, for about $\frac{9}{64}\lambda_\pm^2$ negative pieces, or $L - \frac{1}{2}\lambda_\pm^2$ when the L pieces include positive ones near their corner $t_j = t_{j+1} = -1$ with $(e_j, e_{j+1}) = (1, 0)$, where only $v \cdot U = 0$ is available and $c_1(\mathfrak{s})_P^2$ is unbounded.

Sketch. (a) w is odd on a square-zero class of the old cap, so $v_C \neq 0$, and a generic cap metric avoids the finitely many relevant walls; the bound uses Weitzenböck estimates on unit strips and the decay of harmonic forms. (c) Θ contains $\frac{1}{4}(1 - \lambda^2)$ and $d_s + p \geq 0$ allows $(c_1(\mathfrak{s}) \cdot e)^2 \lesssim 4L + C$, so $|v \cdot e| \gtrsim \lambda - 2\sqrt{L}$ and $i \approx 2L + \frac{3}{2}(v \cdot e)^2 - \frac{1}{2}\lambda^2 \geq -\frac{1}{8}\lambda^2$. $\square$

Broken limits. Without a component in $\mathcal{M}^{*,0}$ they are excluded by Theorem 2.3(a), the least $\sum(q_j + 4) = -2$ over chains without a cap being attained by an isolated P with $(e_j, e_{j+1}) = (0, 1)$, $v \cdot U = 0$, $\kappa = \frac{1}{2}$. With one, interior chains ($\delta \geq -2$) and short cap chains ($\delta \geq 2$) are excluded by (2.2) with (H5e) alone; long cap chains need (H5f): the joint isotropy of a Seiberg–Witten component and a component in $\mathcal{M}^{*,0}$ is the diagonal ± 1 , so single-valued terms coupled to the phase reach the normal cokernel.

Status. VERIFIED: the reducible strata and indices; Proposition 5.1 and its sharpness; Theorem 5.2(a)–(e), (c) also on $8.9 \cdot 10^6$ classes, (e) on $8.5 \cdot 10^5$, (b) against the Christoffel symbols; Theorem 5.3; Proposition 5.4(b), (c) as arithmetic; the counts at broken limits. PLAUSIBLE: Theorem 5.2(f) (65%), the family of [MS, §6] (75%), Proposition 5.4(a) (80–85%), regularity on short end pieces (85%). SPECULATIVE: that the components of Proposition 5.4(c) exist. Confidence: 40%; 55% if they do not.

6. Mixed limits with abelian instantons

Proposition 6.1. Let ζ be a limit with a component in $\mathcal{M}^{*,0}$ and abelian components. (a) The abelian components contain no cap, so $k \geq 2$. (b) The stabilizer of ζ in $\prod_j \mathcal{G}_j \times S^1$ is $\{\pm 1\} \times \prod_{\text{asd}} \{\pm 1\} \times \prod_{\text{abelian}} \text{U}(1)$, with $\text{SU}(2)$ for $v = 0$. (c) At an abelian component with $\kappa_0 = -\frac{1}{4}v^2$ the deformation complex splits under $\text{U}(1)_b \times S^1$ into complexes of weights $(0, 0)$, $(2, 0)$, $(1, 1)$, $(-1, 1)$ and indices $p - (1 + b^+)$ (real), $4\kappa_0 - (1 + b^+)$, $\frac{1}{8}((\Lambda + v)^2 - \sigma)$, $\frac{1}{8}((\Lambda - v)^2 - \sigma)$ (complex), the last two summing to $\Theta - \kappa_0$. (d) On a union of pieces with $b^+ = 0$ every class $v \equiv w$ carries an abelian instanton for every metric and equivariant perturbation, no equivariant perturbation changes the indices of nonzero weight, and $\text{U}(1)_b$ is not fixed by the phase of a component in $\mathcal{M}^{*,0}$. (e) $\text{U}(1)_b$ acts freely on the gluing parameters at the two adjacent necks, with quotient $S^2 \times \text{SU}(2)$.

So abelian instantons can be neither removed nor made regular, as [FS] says. They need not be:

Theorem 6.2. Assume (H5). A limit with a stretched three-sphere, a component in $\mathcal{M}^{*,0}$, abelian components and no component of low index has $D_F \leq -1$; no regularity, gluing or obstruction statement about the abelian components is used. A chain of reducible components without a cap, abelian, Seiberg–Witten or mixed, has $\delta \geq -2$ when the positive pieces are at distance at least two, with equality, at distance at least three, only for a positive piece alone between two stretched three-spheres with $(e_j, e_{j+1}) = (0, 1)$, $\kappa = \frac{1}{2}$, $v \cdot U = 0$, carrying V_{S_j} and $V_{S_{j+1}}$.

Sketch. Theorem 2.3(b) uses only the numbers i_j outside F . For a chain of L pieces, m of them positive, carrying s constraints, $\delta = 6\kappa_\Gamma - m + L - 2s + (e_a - e_b)$, e_a, e_b at its ends, and the bound $\kappa_\Gamma \geq \frac{1}{2}(s^- + m)$ of Proposition 5.1 and Theorem 5.2 on the chain gives the claim; exact minimization over all chains at distances 2 to 5 confirms it. $\square$

At a cap an abelian component would have $i \approx -\frac{1}{2}\lambda^2$ and could not be made regular; Proposition 5.4(a) excludes it. The bound $D_F \leq -1$ is attained only by two components in $\mathcal{M}^{*,0}$ containing the caps, around the isolated P of Theorem 6.2; its Seiberg–Witten version has $c_1(\mathfrak{g})^2 = -4$, so $d_s = -2$ with no parameter, and its abelian version needs $v \cdot \omega = 0$ at the fixed corner period, so tangential regularity and a generic corner metric give a second dimension to spare, away from cap anti-self-dual components of index -1 . So the transversality “after forgetting the abelian instantons” of [FS, §4] is not needed, and the open point named there is misplaced. No choice of w , Λ or holonomy perturbation removes abelian instantons; an Euler-class count would need an obstructed gluing theorem in families with faces [FLM, Hyps. 7.8.1, 11.3.5]; and further blow-ups of the caps cannot raise the least indices of anti-self-dual and Seiberg–Witten cap components together.

Status. VERIFIED: Proposition 6.1, also on 63 410 configurations; Theorem 6.2, with exact chain minima; the statement on blow-ups. PLAUSIBLE: Proposition 5.4(a) (85%); the second dimension (75%, not used). Confidence, given the inputs of §§2, 5: 90% that abelian instantons need no analytic hypothesis, 85% that these limits are excluded.

7. The hypothesis (H5), revised

(H5) of [FS] consists of the following, for the family of the Introduction, the composites Π_M and every $R \geq \max(R_0(M), T_0(M))$.

(H5a) *Order of choices; compactness.* Choose in order the cap metrics and C_W , the odd $\lambda_{\pm}$, Π_0 and p' , M , the scale ϵ of the tubes with the regularization and isolation lengths of the positive pieces, R , then the other perturbations, generically; the closure of $\mathcal{Z}_e$ over $\bar{Q}$ is Uhlenbeck compact, with the bounds of Proposition 1.1 and Lemma 4.2. (75%)

(H5b) *The family.* Round necks along the J_i ; on $\hat{\nu}_i$ the metric of Lemma 3.1 and anti-self-dual determinant connections, the same for both lifts; on the blocks the toric family and tube of Theorem 5.2; the composition law [FS, Thm. 3.1] for this family. (70%)

(H5c) *Control of reducibles.* Proposition 5.4(a) and Theorem 5.2(f), with constants independent of R . (65%)

(H5d) *Representatives.* The transversality term of V_{S_i} is continuous for Uhlenbeck convergence with no point of concentration on S_i and vanishes near reducible restrictions with $\langle v, S_i \rangle = 0$; at $t_i = -1$, V_{S_i} degenerates to the divisors of the capped halves, transverse to the moduli spaces of the completed halves; at $t_i = +1$ it is the same for both lifts. (85%)

(H5e) *Regularity of single components*, by perturbations acting on one piece at a time: the strata in $\mathcal{M}^{*,0}$ at every level, with their constraints, parameters and the μ_c -sections on products; anti-self-dual components; Seiberg–Witten components for their tangential problem. (85%)

(H5f) *Transversality of the product problem in the end regions*, through the phase of the spinor on another piece: two-valued terms in the Dirac equation at anti-self-dual components of low index, single-valued terms in the normal equations at long Seiberg–Witten cap components; compatible with Uhlenbeck compactness, the links, the lens collars and the equality of the data for e , $e + e_i$ off νS_i , with Stokes’ theorem for the resulting weighted branched one-manifold. New, without precedent for $\mathrm{SO}(3)$ -monopoles. (55–60%)

(H5g) [FL2b, Prop. 3.29] over $\bar{Q}$, and gluing at the reducible of energy $\frac{1}{4}$ on $\hat{\nu}_i$, one end per exterior solution, oriented as in Theorem 3.3. (85%, 80%)

These replace transversality at all broken limits and after forgetting abelian instantons, and “uniformly in the neck length”; in [FS, Thm. 4.3], $R \geq \max(R_0(M), T_0(M))$, and Proposition 5.1 replaces [FS, Prop. 4.2].

Verdict

Given (H5) as revised, the closure of $\mathcal{Z}_e$ over $\bar{Q}$ consists of $\mathcal{Z}_e$, the 2^{n_a-1} ends at each cut-down instanton (Proposition 1.2) and the ends over the open faces $\{t_i = +1\}$ (Proposition 3.2), which cancel in $\sum_e \epsilon(e)$ (Theorem 3.3); nothing else occurs (Theorems 2.3, 5.3, 6.2), and no neck along $Y_{\pm 2}$ is stretched. So [FS, Thm. 4.1] gives $2^{n_a-1}\Omega(X(\Pi_M)) = 0$, hence $\Omega(X(\Pi_M)) = 0$, for all large M (VERIFIED as logic), with nothing glued at a reducible. The main analytic difficulty is (H5f), already at faces of codimension one; abelian instantons play no part in it, and its part at long Seiberg–Witten cap components may be empty (SPECULATIVE).

Confidence: counts and lattice arithmetic 97%; the six groups of strata 80%, 45%, 75%, 70%, 40%, 85% (the last given the shared inputs). (H5) as a whole is SPECULATIVE, at about 25% (15–35%), below the 30% of [FS], as (H5f) now includes the long Seiberg–Witten cap components and the composition law must hold for the block family, which outweighs dropping the hypothesis at abelian instantons; so about 25% that $\Omega(X(\Pi_M)) = 0$ for large M by this argument. As (H1), (H2) and (H4) are known, and (H3) is at 85–90%, the cosmetic conclusion by this argument is SPECULATIVE, at about 20% (12–30%).

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